

Class 8: Mini Project

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Background

Today we will learn about methods and technique about clustering and PCA

```
url <- "https://bioboot.github.io/bimm143_W26/class-material/WisconsinCancer.csv"
wisc.df <- read.csv((url))
wisc.data <- read.csv(url, row.names = 1)
```

Have a wee peak at the first few rows

```
wisc.data <- wisc.data[,-1]
head(wisc.data)
```

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
842302	17.99	10.38	122.80	1001.0	0.11840
842517	20.57	17.77	132.90	1326.0	0.08474
84300903	19.69	21.25	130.00	1203.0	0.10960
84348301	11.42	20.38	77.58	386.1	0.14250
84358402	20.29	14.34	135.10	1297.0	0.10030
843786	12.45	15.70	82.57	477.1	0.12780
	compactness_mean	concavity_mean	concave.points_mean	symmetry_mean	

842302	0.27760	0.3001	0.14710	0.2419
842517	0.07864	0.0869	0.07017	0.1812
84300903	0.15990	0.1974	0.12790	0.2069
84348301	0.28390	0.2414	0.10520	0.2597
84358402	0.13280	0.1980	0.10430	0.1809
843786	0.17000	0.1578	0.08089	0.2087
fractal_dimension_mean radius_se texture_se perimeter_se area_se				
842302	0.07871	1.0950	0.9053	8.589 153.40
842517	0.05667	0.5435	0.7339	3.398 74.08
84300903	0.05999	0.7456	0.7869	4.585 94.03
84348301	0.09744	0.4956	1.1560	3.445 27.23
84358402	0.05883	0.7572	0.7813	5.438 94.44
843786	0.07613	0.3345	0.8902	2.217 27.19
smoothness_se compactness_se concavity_se concave.points_se				
842302	0.006399	0.04904	0.05373	0.01587
842517	0.005225	0.01308	0.01860	0.01340
84300903	0.006150	0.04006	0.03832	0.02058
84348301	0.009110	0.07458	0.05661	0.01867
84358402	0.011490	0.02461	0.05688	0.01885
843786	0.007510	0.03345	0.03672	0.01137
symmetry_se fractal_dimension_se radius_worst texture_worst				
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
84358402	0.01756	0.005115	22.54	16.67
843786	0.02165	0.005082	15.47	23.75
perimeter_worst area_worst smoothness_worst compactness_worst				
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249
concavity_worst concave.points_worst symmetry_worst				
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
84358402	0.4000	0.1625	0.2364	
843786	0.5355	0.1741	0.3985	
fractal_dimension_worst				
842302	0.11890			

842517	0.08902
84300903	0.08758
84348301	0.17300
84358402	0.07678
843786	0.12440

Make sure to remove the first `diagnosis` column- - I don't want to use this for my machine learning models. We will use it later to compare our results to the expert diagnosis.

```
diagnosis <- wisc.df$diagnosis
```

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

```
ncol(wisc.data)
```

```
[1] 30
```

Q2. How many of the observations have a malignant diagnosis?

```
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

```
table(wisc.df$diagnosis)
```

```
  B   M  
357 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
column_names <- colnames(wisc.data)  
length(grep("_mean", column_names))
```

```
[1] 10
```

Principle Component Analysis

We want to scale and center our data before PCA to make sure that each component have equal contribution. The main function here is `prcomp()` and we want to make sure we set the optional argument `scale=TRUE`:

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335

	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966

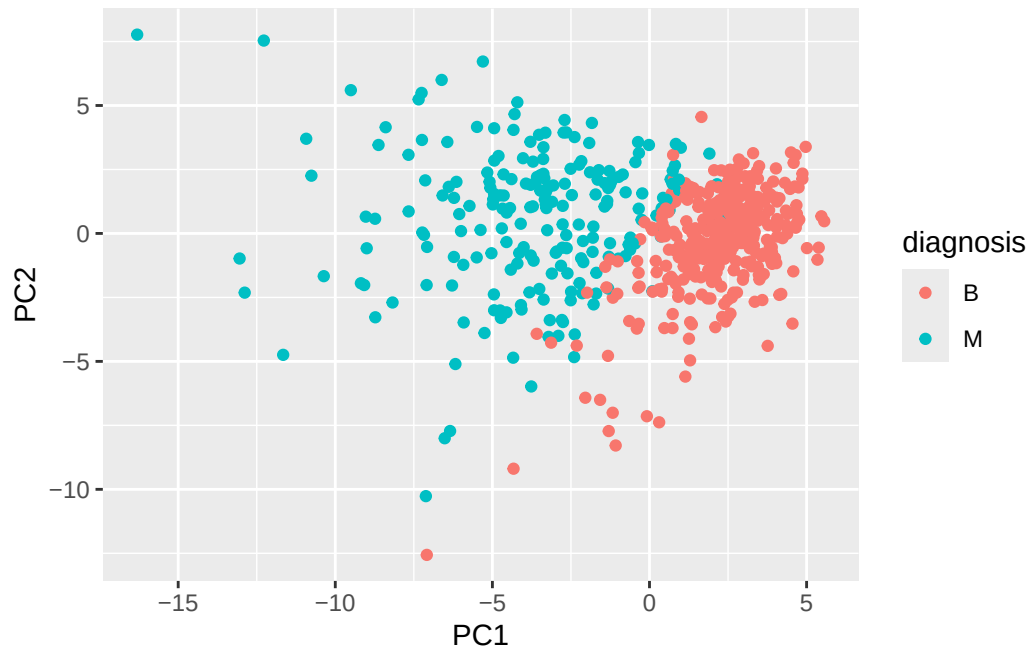
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997

	PC29	PC30
Standard deviation	0.02736	0.01153
Proportion of Variance	0.00002	0.00000
Cumulative Proportion	1.00000	1.00000

Our main PCA “score plot” or “PC plot” of results:

```
library(ggplot2)
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



```
data.scaled <- scale(wisc.data)
```

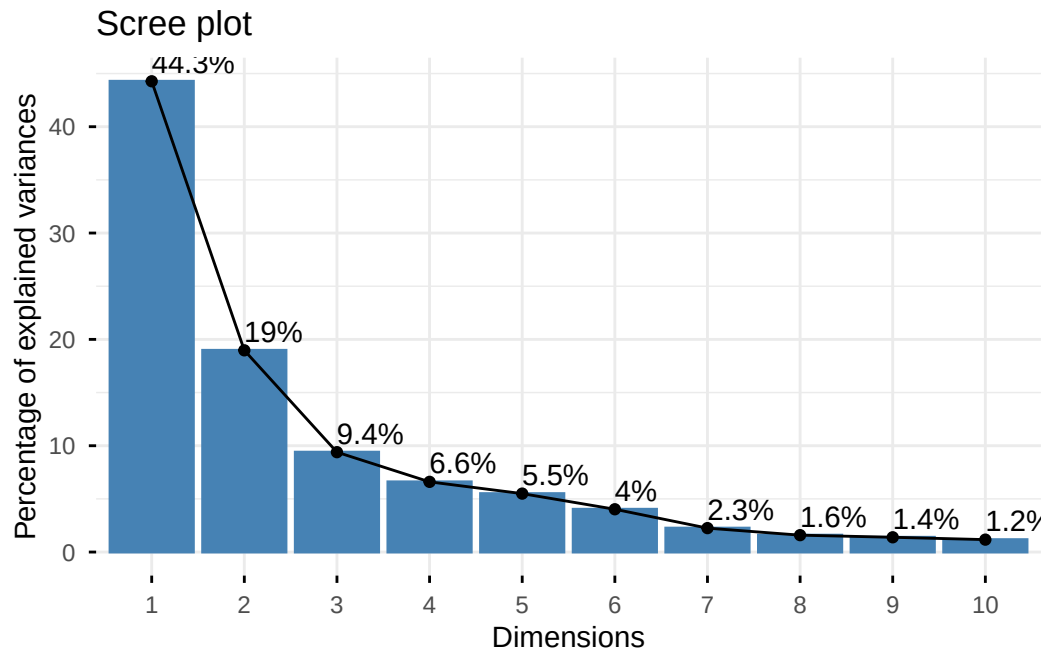
Score_plot:

```
## ggplot based graph
#install.packages("factoextra")
library(factoextra)
```

Welcome! Want to learn more? See two factoextra-related books at <https://goo.gl/ve3WBa>

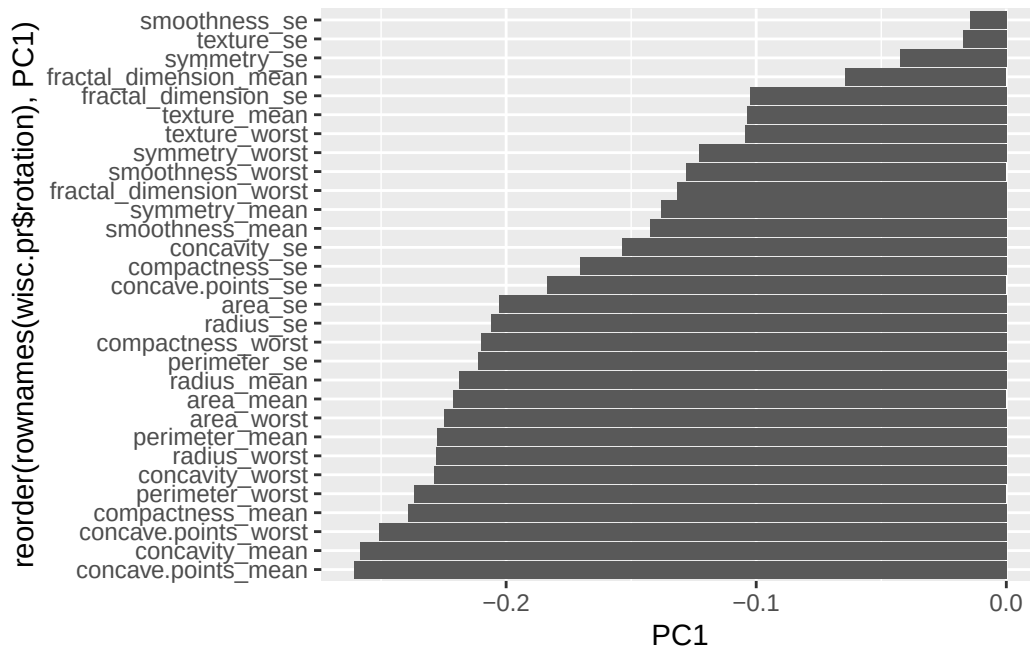
```
fviz_eig(wisc.pr, addlabels = TRUE)
```

Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
Ignoring empty aesthetic: `width`.



The loading vector (`wisc.pr$rotation`) tells us which original measurements contribute most to each PC.

```
ggplot(wisc.pr$rotation)+  
  aes(PC1,  
       reorder(rownames(wisc.pr$rotation), PC1)) +  
  geom_col()
```



Collectively these two plots (“score plot” and “loading plot”) tell us that if cell nucleus are deeply indented (“concave”), irregular non circular (“compactness”), and have large “perimeter” values they tend to be malignant...

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

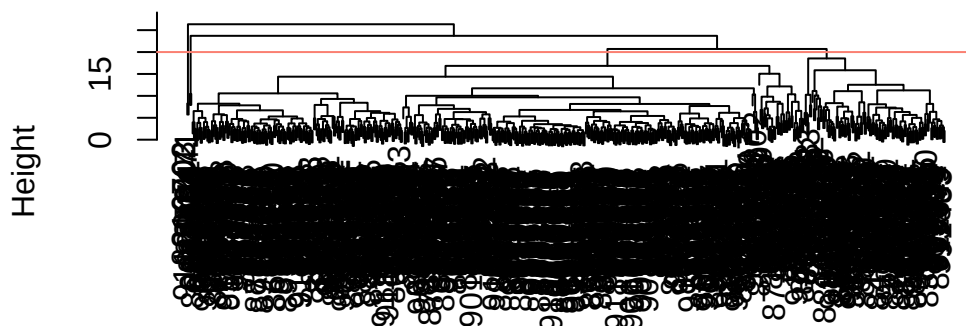
```
[1] -0.2608538
```

Hierarchical Clustering

First scale the data (`scale`), then calculate a distance matrix (with `dist()` function). Then cluster with the `hclust()` function and plot

```
wisc.hclust <- hclust(dist(scale(wisc.data[, -1])))
plot(wisc.hclust)
abline(h = 20, col="salmon")
```

Cluster Dendrogram



```
dist(scale(wisc.data[, -1]))  
hclust(*, "complete")
```

You can also use `cutree()` function with an argument `k=4` rather than `h=height`

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)  
table(wisc.hclust.clusters)
```

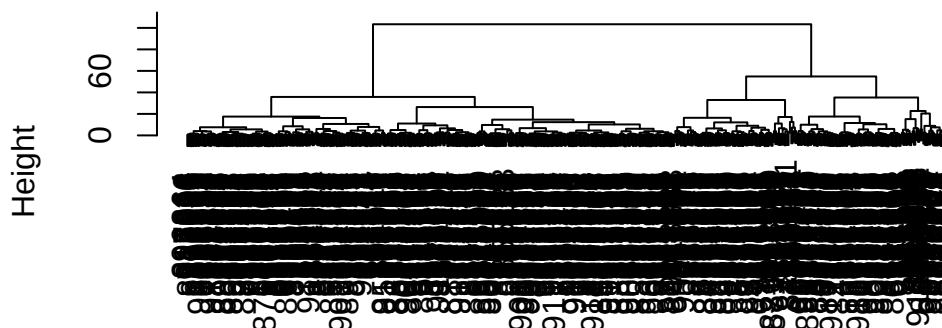
```
wisc.hclust.clusters  
  1  2  3  4  
106 459  2  2
```

Combining methods

Here we will take our PCA results and use those as input for clustering. In other words our `wisc.pr$x` scores that we plotted above (the main output from PCA - how the data lie on our new principle component axis/variables) and use a subset of the PCs as input for `hclust()`

```
pc.dist <- dist(wisc.pr$x[, 1:3])  
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")  
plot(wisc.pr.hclust)
```


Cluster Dendrogram



```
pc.dist
hclust (*, "ward.D2")
```

Cut the dendrogram/tree into two main groups/clusters:

```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1  2
203 366
```

I want to know how the cluster in `grps` with values of 1 or 2 correspond the expert `diagnosis`

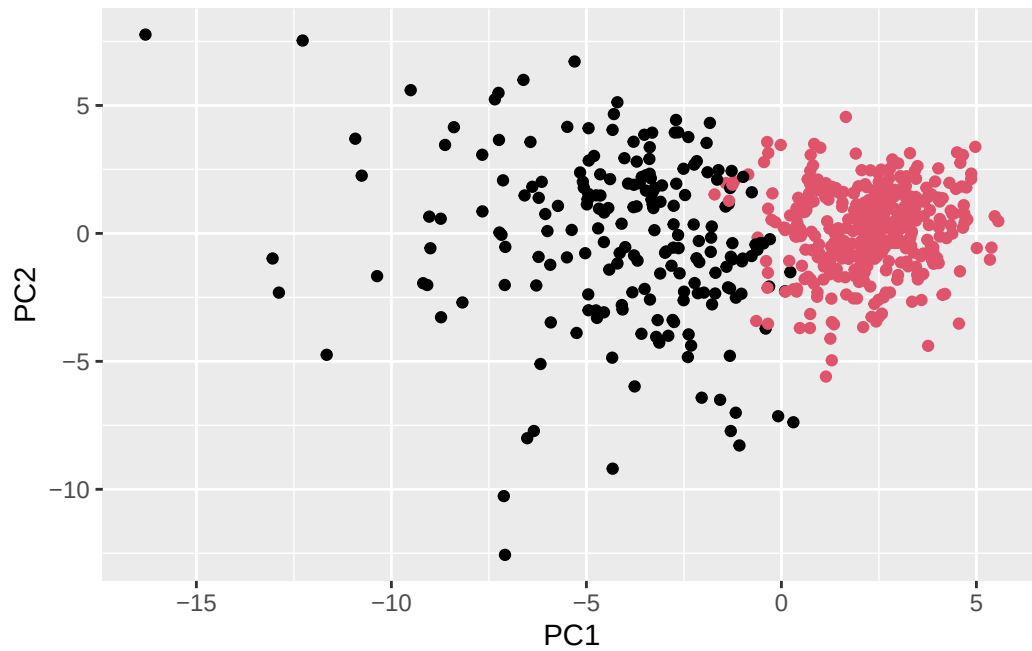
```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1   24 179
  2 333  33
```

My clustering **group 1** are mostly “M” diagnosis (179) and my cluster **group 2** are mostly “B” diagnosis (333)

24 False positive (FP) 179 True positive (TP) 333 True negative (TN) 33 False negative (FN)

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC2) +  
  geom_point(col=grps)
```



Sensitivity

$TP / (TP + FN)$

```
179 / (179 + 33)
```

```
[1] 0.8443396
```

$TN / (TN + FP)$

```
333 / (333 + 24)
```

```
[1] 0.9327731
```

Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```

